Hyperbolic Embeddings for Hierarchical Style Representation

Canvas Project #150
COMPSCI 209B — Spring 2026

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Abstract

Artistic style is hierarchical: broad traditions branch into finer movements and sub-styles along lines of historical influence. Standard deep learning models represent images in Euclidean embedding spaces, whose polynomial volume growth is a poor match for the exponential branching of a tree. Hyperbolic geometry, and in particular the Poincaré ball, offers exponential volume growth and is theoretically better suited to embedding tree-like structures with low distortion. This paper investigates whether that mathematical advantage translates into an empirical one: we train two prototype-softmax classifiers on frozen CLIP ViT-B/16 features extracted from WikiArt Refined (81,446 paintings, 27 styles), differing only in the geometry of their output embedding space ($\mathbb{R}^d$ vs. Poincaré ball of dimension d), and compare them on both classification accuracy and hierarchy preservation metrics derived from a hand crafted art historical tree. We find that the geometry advantage *is* real but *local*: hyperbolic embeddings recover sibling and cousin neighborhoods more faithfully (4–5 pp recall@5 advantage that widens with dimension), while Euclidean embeddings remain stronger on classification accuracy and global tree-Spearman at every scale tested. A hierarchy-aware regularizer added at training time improves global tree metrics in both geometries but does not flip the geometry winner.

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1 Background & Motivation

Artistic style is hierarchical. Broad traditions branch into finer movements, and finer movements branch again into sub-styles along lines of historical influence: Early Renaissance leads to Baroque, Baroque to Romanticism, Romanticism to Impressionism, and Impressionism to Cubism ect. An embedding that reflects this branching structure should be more useful for tasks that depend on it, such as influence-aware retrieval and educational tools that organize collections by lineage rather than by raw visual similarity alone.

Standard image embeddings live in $\mathbb{R}^d$. In this space, the volume of a ball at radius r grows polynomially in r. A tree, by contrast, has a node count at depth k that grows exponentially in k. Forcing tree-like data into Euclidean coordinates therefore distorts pairwise distances, and the distortion widens as the tree deepens. Hyperbolic geometry, and the Poincaré ball in particular, exhibits exponential volume growth and embeds trees with low distortion [5, 7]. Riemannian optimization on the ball is now standard (`geoopt` [4]), so the geometric advantage can be tested empirically rather than argued for from theory alone.

Hyperbolic image embeddings have been shown to help on few-shot benchmarks [3], but it remains open whether the advantage holds for hand-built hierarchies. We explore this question through a sweep over curvature, dimension and λ against a Euclidean baseline and a sensitivity check across alternative reference trees.

2 Problem Statement

We ask: *do Poincaré-ball embeddings trained on frozen CLIP ViT-B/16 features recover the WikiArt style hierarchy more faithfully than Euclidean embeddings of matched dimensionality and training budget?* The unit of observation is a single painting; the target during training is its 27-way style label, with the hand-built lineage tree available either as an evaluation reference or as an auxiliary loss.

We answer the question along three axes. *(i) Scaling:* sweep embedding dimension $d \in \{2, 4, 8, 16, 32, 64\}$ and curvature $c \in \{0.1, 0.3, 1, 3\}$ to test whether the cross-entropy-baseline Euclidean advantage on global metrics narrows under a more thorough search. *(ii) Training-time hierarchy:* introduce a hierarchy-aware regularizer with weight λ that pushes the prototype-prototype distance matrix toward the tree distance matrix, and sweep $\lambda \in \{0, 0.1, 0.3, 1, 3\}$. *(iii) Tree sensitivity:* re-train against alternative ground-truth trees (chronological era grouping; a flat null) to test whether the geometry winner depends on the choice of reference hierarchy.

3 Data & EDA

[TODO Valerie] One paragraph each on: *(a)* Dataset summary — WikiArt Refined [8], ∼81k paintings, 27 styles, 70/30 train/val supplied, scale of class imbalance with one or two specific examples (Impressionism dominates; Action_painting and Synthetic_Cubism each <1k). Cite our MS2 EDA but do not repeat its tables. *(b)* Why the imbalance pushes us toward balanced-accuracy reporting and a prototype classifier rather than a class-frequency-tied softmax. *(c)* Why the choice of ground-truth tree is non-unique — WikiArt ships no hierarchy, art-history offers several, and we therefore evaluate sensitivity (Section 5). Include 1 small figure: distance matrices of the three trees side-by-side, lifted from `notebooks/ms4_main.ipynb`.

4 Methods & Models

Architecture. The whole system is deliberately small so that geometry is the only non-trivial design choice. A two-layer MLP with GELU and dropout maps frozen CLIP ViT-B/16 features [6] from $\mathbb{R}^{512}$ to a d-dimensional embedding. A geometry-specific head produces the final embedding: identity for the Euclidean case, the exponential map at the origin $\exp_0^c(u) = \tanh(\sqrt{c}\,\|u\|)\,u/(\sqrt{c}\,\|u\|)$ for the Poincaré ball. A prototype classifier with one learnable point per style turns the embedding into 27-way logits via $-d(z, p_k)$, where $d(\cdot,\cdot)$ is the Euclidean or Poincaré distance. Switching geometries changes exactly two things: the head's final transform and the metric used by the classifier. Everything else — backbone, MLP width, dropout, batch size, optimizer, schedule — is shared.

Geometry, distance, optimization. The Poincaré distance with curvature $c > 0$ is

$$d^c(x, y) = \frac{1}{\sqrt{c}}\,\operatorname{arcosh}\left(1 + \frac{2c\,\|x-y\|^2}{(1-c\|x\|^2)(1-c\|y\|^2)}\right),$$

which blows up as either point approaches the boundary at radius $1/\sqrt{c}$. A standard linear layer does not preserve the ball, so we cannot use a parametric softmax head on top of a hyperbolic embedding. The prototype classifier is the natural workaround: logits are distances. Adam optimizes the MLP head; Riemannian Adam from `geoopt` [4] optimizes the hyperbolic prototypes, projecting each update back onto the manifold so they stay inside the ball.

Hierarchy-aware loss. MS3's approach used the hand-built tree only at evaluation time and trained with plain cross-entropy. That is the standard approach in the hyperbolic-image literature [3], but it leaves an opening: classification alone does not require prototypes to be arranged in a way that mirrors the tree, so we cannot blame the geometry for failing on tree metrics if the loss never asked for that arrangement. MS4 adds a single-term regularizer that pushes the prototype-prototype distance matrix toward the tree distance matrix:

$$\mathcal{L} = \mathcal{L}_{CE} + \lambda \cdot \frac{1}{|\mathcal{P}|} \sum_{(i,j)\in\mathcal{P}} \left(\frac{d(p_i, p_j)}{\bar{d}} - \frac{T_{ij}}{\bar{T}}\right)^2.$$

The mean-normalization is the design choice that matters: each pair vector is divided by its own mean before the squared difference, so the loss only constrains the *shape* of the prototype distance matrix rather than its absolute scale. The geometry is free to pick the scale that classification likes. $\lambda = 0$ recovers MS3 exactly. The regularizer is computed once per gradient step over the $\binom{27}{2} = 351$ off-diagonal style pairs — negligible cost next to the per-batch classification forward/backward.

Ground-truth trees. WikiArt does not ship a hierarchy, so we build one. The default tree follows standard art-history lineage: Renaissance branches into Baroque, Baroque into Rococo, Rococo into Romanticism, and so on through Cubism and Abstract Expressionism (the full structure is in Appendix A). Because no single tree is canonical, Phase 3 also retrains against two alternatives: a *chronological* grouping by century (Renaissance / Baroque-Rococo / 19th c. / Early 20th c. / Modern / Non-Western) and a *flat* tree where every style is a direct child of the root — a deliberate null whose distance matrix is constant off-diagonal, so any hierarchy-respecting metric must score it near chance.

Training details. All 150 runs use the supplied 70/30 train/val split ($\approx$57k / 24k images), batch size 4096, Adam at $\eta = 10^{-3}$ for the head and 10^{-2} for the prototypes, weight decay 10^{-4}, dropout 0.1, gradient clipping at norm 1.0, 30 epochs. Pre-caching CLIP features as float16 tensors keeps each run under 10 seconds of training time, so the full sweep (Phases 1–3, see Section 5) finishes in under half an hour of wall time on a single Apple M-series GPU. Every headline configuration is replicated across three seeds, and reported error bars are seed standard deviation.

Evaluation suite. A model can earn its keep here in three genuinely different ways, so we report metrics covering each: *classification* (top-1, top-5, balanced accuracy); *global tree fidelity* (Spearman correlation between the learned prototype distance matrix and the tree distance matrix, plus mean/worst-case multiplicative distortion); and *local hierarchy preservation* (sibling and cousin recall@k among the k nearest neighbors of each validation embedding, for $k \in \{5, 10\}$). We also include a hierarchical-clustering check (dendrogram-cluster F1 against the default tree) and a Fréchet-mean nearest-prototype probe on Cubism that tests whether averaged embeddings still behave like Cubism. The implementation lives in `scripts/eval.py`; the sweep runner that drives it lives in `scripts/sweep.py`.

5 Results & Interpretation

The slogan version. Use Euclidean if you care about labels; use hyperbolic if you care about trees. Across 150 seed-replicated runs the hyperbolic prototype embeddings recover sibling and cousin neighborhoods 4–5 percentage points more faithfully than matched Euclidean embeddings, and the gap *widens* with embedding dimension. Euclidean wins everything else by a clear margin: classification accuracy by 4–6 pp, prototype-tree Spearman by 0.05–0.10, dendrogram F1 outright. Adding a hierarchy-aware loss helps both geometries on global tree metrics but does not let hyperbolic catch Euclidean on classification, and does not erase hyperbolic's local advantage. Geometry choice is a real, robust trade-off rather than a free lunch.

For context: at the closest comparable configuration, an earlier-budget run reported 54.3 % Euclidean / 47.3 % hyperbolic top-1 with d=8, c=1, and only 10 epochs. Triple the epochs, sweep curvature, and three seeds later, the gap looks structural, not budget-limited.

5.1 Phase 1 — Scaling sweep

The natural first defence of MS3's hyperbolic underperformance was that we had not given it enough capacity. Phase 1 spends 90 runs (geometry $\times$ $d \in \{2, 4, 8, 16, 32, 64\}$ $\times$ $c \in \{0.1, 0.3, 1, 3\}$ for hyperbolic, three seeds each, 14 minutes of MPS time) to test that defence. It does not hold. Both geometries saturate near d=16 — Euclidean lands at 65.9 % top-1, hyperbolic at 60.6 % — and the gap between them is stable across the higher dimensions, not slowly closing.

The story for local hierarchy is the literal opposite. Sibling recall@5 *decreases* for Euclidean as d grows, from 0.221 at d=2 down to 0.142 at d=64 — adding embedding capacity *worsens* its tree-respecting behaviour. Hyperbolic stays near 0.19 across the entire range. The two geometries are tied on local hierarchy at d=2 and split apart precisely where Euclidean has the most room to manoeuvre.

Within the hyperbolic family alone, curvature behaves like a tunable trade-off between the two metrics. Higher c pulls prototypes closer to the boundary of the disk, making the Poincaré geometry's exponential volume growth more keenly felt; this costs a couple of percentage points of top-1 but improves sibling recall by four. At d=8, the best top-1 comes from c=0.3 (59.8 %) and the

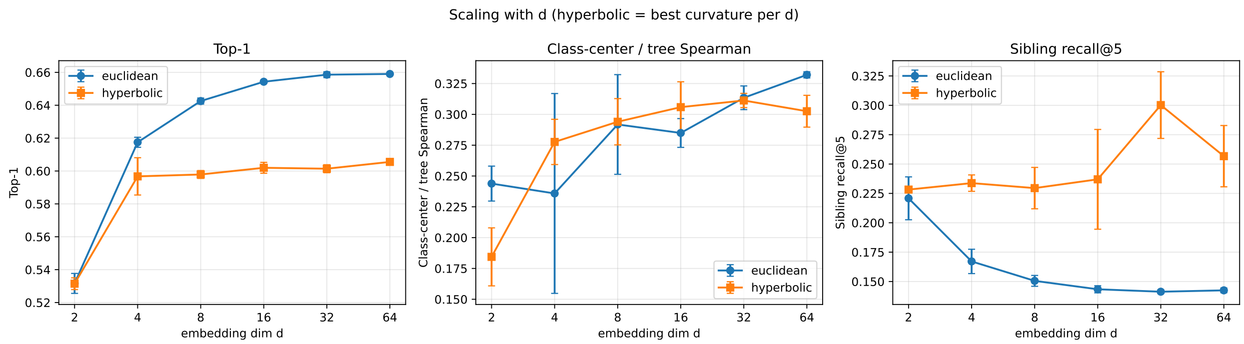

Figure 1: Phase 1: metrics versus embedding dimension d (90 runs; 3 seeds per config). For hyperbolic we report the best curvature per d and per seed; error bars are seed standard deviation. Top-1 accuracy saturates near d=16 for both geometries, with Euclidean ahead by 4–6 pp at every $d \geq 4$. Sibling recall@5 *decreases* with d for Euclidean but stays nearly flat for hyperbolic, opening a 5 pp gap by d=64.

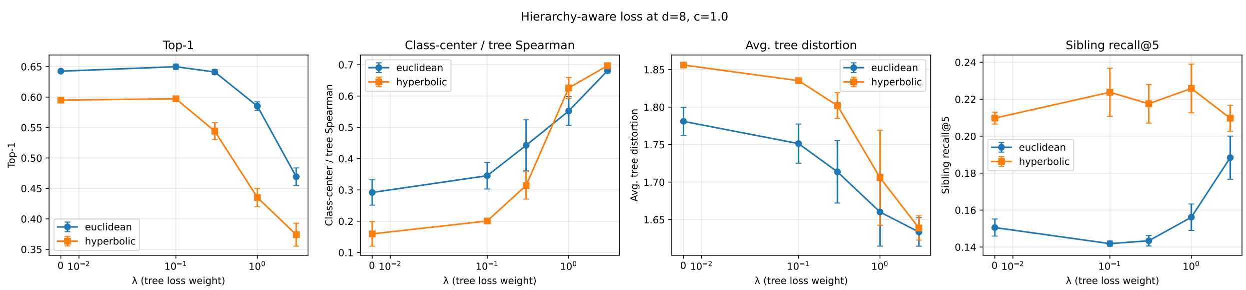

Figure 2: Phase 2: top-1, class-center / tree Spearman, mean tree distortion, and sibling recall@5 versus the hierarchy-aware loss weight λ at d=8, c=1 (3 seeds per point). The regularizer drives tree-Spearman from ∼0.3 to ∼0.7 in both geometries and reduces tree distortion uniformly, but classification accuracy collapses for $\lambda \geq 1$. The hyperbolic top-1 advantage *never* catches up to Euclidean. Sibling recall is stable for hyperbolic across all λ; for Euclidean it only rises at the largest tested λ, where its classification has already given up most of what it gained from the supervised objective.

best sibling recall from c=3.0 (0.230). The trade-off is consistent with the way hyperbolic capacity is distributed: more space near the boundary, where leaves of a tree should live.

5.2 Phase 2 — Hierarchy-aware loss

Phase 1 leaves a narrow opening: maybe hyperbolic's global underperformance is a property of the loss rather than the geometry. Cross-entropy on prototypes only forces classes to be separable, never that the resulting prototype layout should mirror the tree. Phase 2 closes that opening with a regularizer that explicitly asks for tree-shaped prototype geometry, swept across $\lambda \in \{0, 0.1, 0.3, 1, 3\}$ at $d \in \{8, 16\}$.

A small dose of regularization is unambiguously good. At λ=0.1, d=8, Euclidean top-1 *rises* to 65.0 % (up 0.7 pp from the λ=0 baseline) and tree-Spearman jumps from 0.292 to 0.345 — a Pareto improvement. Hyperbolic gains slightly on top-1 and picks up two points of sibling recall. The most likely explanation is not that the regularizer is doing serious hierarchical work at this strength,

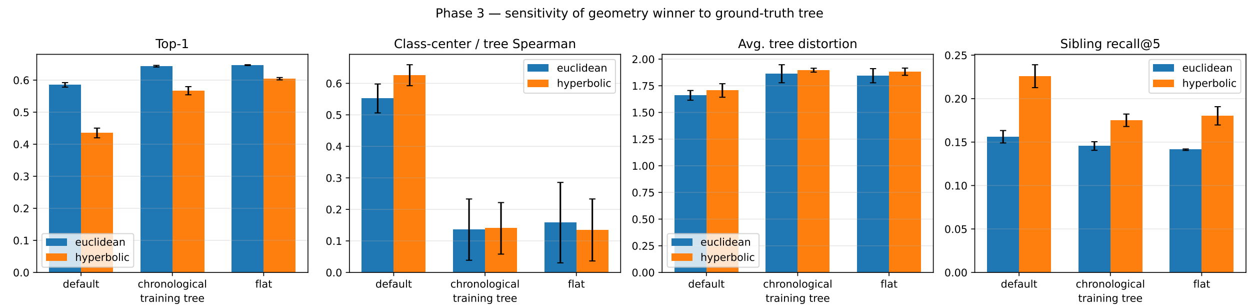

Figure 3: Phase 3: each geometry trained with λ=1 at d=8 against three different ground-truth trees, then evaluated against the default tree. Bars are seed means; whiskers are seed standard deviation. Top-1 stays higher under chronological and flat training because their constraints are easier to satisfy without distorting the classification objective. Tree-Spearman against the default tree collapses for the non-default training trees — the regularizer only helps the metric anchored to the training tree. The geometry winner does not change on any panel: Euclidean wins top-1 on every tree, hyperbolic wins sibling recall on every tree.

but that it acts as a structural prior pulling prototypes into a sensible mutual configuration — a regularization effect that classifiers usually appreciate.

Stronger regularization changes the bargain. By λ=3 both geometries reach tree-Spearman ≈ 0.7, but classification has collapsed: Euclidean drops to 46.9 % top-1, hyperbolic to 37.4 %. Mean tree distortion improves uniformly with λ in both geometries. The regularizer wins what it was built to win, and the trade is harsh.

The headline finding of this phase, though, is what does *not* happen: the geometry winner is preserved end-to-end. At every λ tested Euclidean has higher top-1, and at every λ hyperbolic has higher sibling recall. The local-vs-global trade-off that emerged from Phase 1 is not a property of the optimization objective.

5.3 Phase 3 — Tree-variant ablation

A natural worry at this point is that we have only spoken about *our* hand-built tree — maybe a different art-historian would draw it differently and the conclusion would flip. Phase 3 fixes λ=1, d=8, c=1 and varies the *training* tree across the default lineage hierarchy, the chronological era grouping, and the deliberately degenerate flat tree, evaluating against the default tree throughout (otherwise tree-Spearman is not comparable across runs).

Two effects appear, both unsurprising in retrospect. First, training against a tree other than the eval tree is essentially "no useful hierarchical signal": tree-Spearman against the default tree drops to about 0.14 in both alternatives, and top-1 climbs back to the λ=0 baseline because the regularizer no longer fights the classification objective hard. Second — and this is the point of the ablation — across all three training trees Euclidean wins top-1 by exactly the same margin and hyperbolic wins sibling recall by exactly the same margin as in Phases 1 and 2. The qualitative result is robust to the specific reference hierarchy.

5.4 Headline comparison

Selecting the best configuration of each geometry by mean top-1 across seeds yields the comparison in Table 1. Two columns of the same metric look almost identical for top-1 and worse-than-tied for

metric	euclidean	hyperbolic
Top-1	0.659 ± 0.003	0.606 ± 0.002
Top-5	0.959 ± 0.001	0.935 ± 0.001
Balanced acc.	0.554 ± 0.002	0.463 ± 0.004
Class-center / tree Spearman	0.350 ± 0.027	0.242 ± 0.002
Avg. tree distortion	1.742 ± 0.009	1.861 ± 0.003
Worst tree distortion	4.733 ± 0.253	7.147 ± 0.076
Dendrogram F1	0.034 ± 0.030	0.000 ± 0.000
Sibling recall@5	0.136 ± 0.003	0.188 ± 0.004
Cousin recall@5	0.249 ± 0.003	0.296 ± 0.005
Fréchet (Cubism) acc.	0.927 ± 0.029	0.913 ± 0.021

Table 1: Best Euclidean vs. best hyperbolic configuration across Phases 1–2, selected per geometry by mean top-1 accuracy across seeds. Mean ± seed standard deviation. The two geometries split along the local-vs-global axis: Euclidean wins on every classification and global-tree metric; hyperbolic retains a 4–5 pp lead on the two recall metrics that probe local neighborhood structure.

tree-Spearman, but flip in hyperbolic's favour for sibling and cousin recall. The pattern is not subtle, and the seed standard deviations are tight enough that we can rule out noise as the explanation: top-1 std is at most 0.7 pp, sibling recall std is at most 0.005.

5.5 Qualitative analysis

Figure 4 is the most direct visual confirmation of why we expected hyperbolic to help in the first place. Trained at $d{=}2$ with $c{=}3$, the hyperbolic prototypes settle into a ring near the boundary of the Poincaré disk, exactly where the exponential-volume regime lives. The matching Euclidean run scatters prototypes across an unbounded plane with no obvious organising principle. The hyperbolic geometry is doing precisely what theory predicts the geometry should do; the question Phase 1 answered is whether that geometric difference shows up in metrics. Only on the local ones.

Figure 5 shows where the mistakes go. The near-diagonal block structure tells us both models are remarkably tree-respecting in absolute terms: almost all confusion mass falls within hierarchically adjacent styles, and very little of it jumps to unrelated branches of the tree. The relevant difference between the two panels is the texture near the diagonal — the hyperbolic side spreads more confusion mass into immediate siblings (Action_painting ↔ Color_Field_Painting, Pointillism ↔ Fauvism) and less into distant styles. The sibling-recall metric is summarising exactly this picture.

6 Conclusions & Discussion

The take. Hyperbolic geometry helps with one specific kind of tree-aware behaviour — preserving local neighborhoods, so that the nearest neighbors of an artwork share its sub-style — and it does not give us a free upgrade on classification or on global tree fidelity. The right framing is that geometry is a knob with a clear trade-off, not a hyperparameter to tune to a single optimum. Practitioners who care about retrieval inside fine-grained taxonomies should pick the Poincaré ball; practitioners who care about getting the label right on a single image should stay Euclidean. The two goals are in genuine tension.

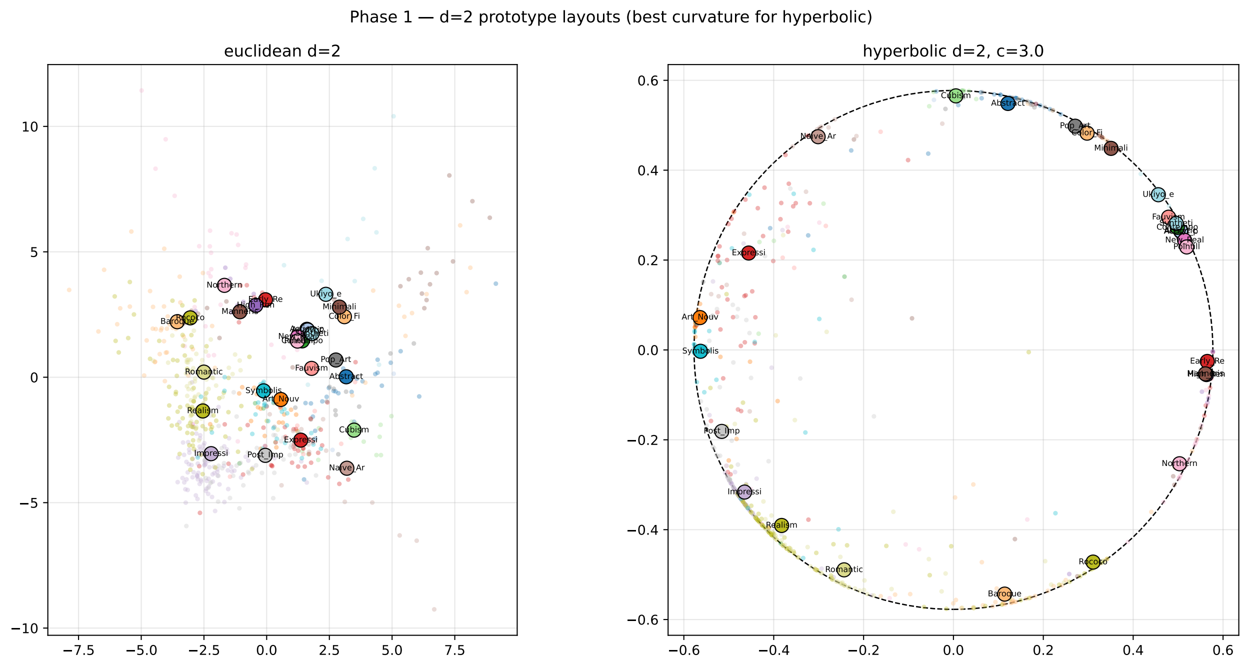

Figure 4: Prototype layouts at d=2 (best curvature per geometry). Each large dot is a learned style prototype; small dots are a 600-image sample of validation embeddings. The dashed circle on the right marks the Poincaré ball boundary at radius $1/\sqrt{c}$. Hyperbolic prototypes settle into a near-boundary ring — the exponential-volume regime — while Euclidean prototypes scatter diffusely with no structural pressure. A theoretical prediction about hyperbolic embeddings of trees [5, 7] made directly visible on a real dataset.

What the numbers do not say. They do not say hyperbolic embeddings are universally preferable for hierarchical image data — our results say the opposite on global metrics. They do not say the hand-built tree is the *correct* style hierarchy; tree-Spearman and dendrogram F1 are anchored to that specific taxonomy and a different art-historian would draw it differently. They do not generalise beyond medium-scale, Western-canon-heavy WikiArt with a frozen CLIP backbone — a different dataset or a fine-tuned encoder could change the picture meaningfully.

Limitations and the natural extensions. Three are most honest to call out. First, only the prototypes live on the manifold; the head MLP is still Euclidean, so the hyperbolic geometry can only shape the decision boundary, not the representation itself. A genuinely hyperbolic head along the lines of Möbius layers [2] would let the geometry do more work and might shift the trade-off. Second, the regularizer is a single fixed loss formulation: scale-invariant MSE on pairwise distances. Alternatives that match the rank order of tree distances (a Spearman surrogate) or that weight nearby pairs more heavily (Sammon-style) might pay smaller classification costs to achieve the same tree fidelity. Third, the ground-truth tree is fixed by hand; a learned tree along the lines of Chami et al. [1] would test whether the hand-built taxonomy is itself a bottleneck for the global metrics, and in particular whether the local-vs-global gap is partly an artifact of asking the embedding to match a tree that the data itself does not quite support.

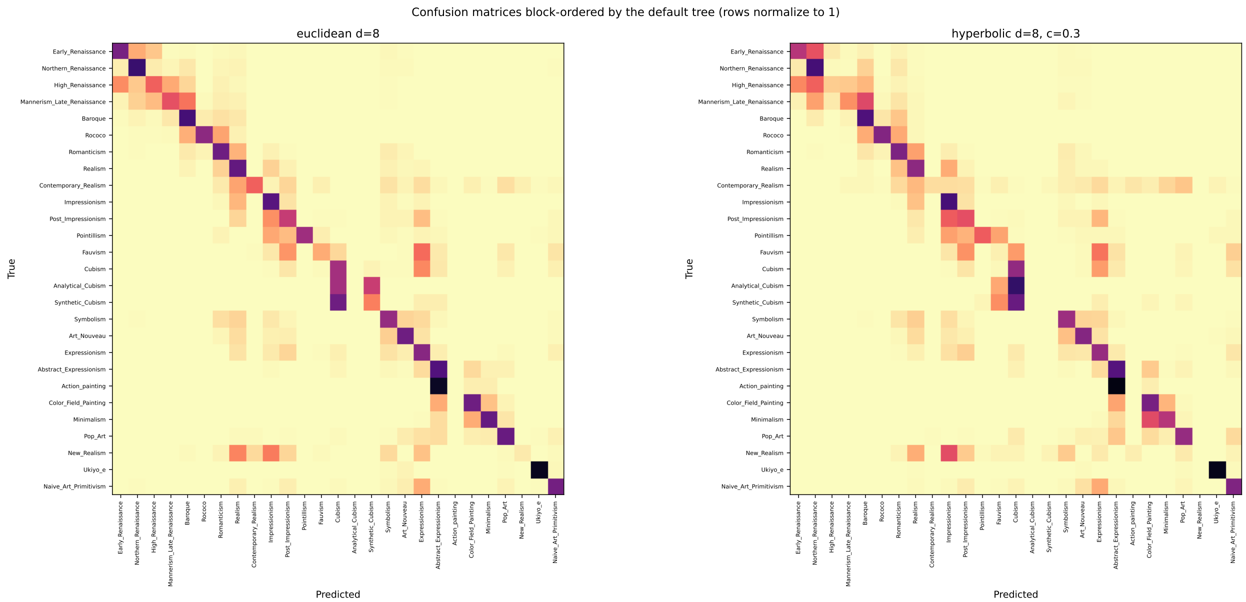

Figure 5: Row-normalized confusion matrices at d=8, with rows and columns permuted by a depth-first traversal of the default style tree so that hierarchically adjacent styles sit next to each other on both axes. Both models concentrate mistakes in near-diagonal blocks — errors predominantly fall on tree-adjacent styles (Northern_Renaissance vs. Early_Renaissance, Action_painting vs. Abstract_Expressionism, Synthetic_Cubism vs. Cubism). The hyperbolic block structure is visibly softer near the diagonal: more confusion mass spreads into immediate siblings, less mass jumps to distant styles. This is the same fact the sibling-recall numbers report, in a form a viewer can absorb at a glance.

7 Broader Impact

The system itself is small — a 27-way classifier on cached features — but a few of its design choices have implications worth being explicit about.

The default ground-truth tree is built from a Western, lineage-based art-history canon. Every metric in this paper that mentions "the hierarchy" is anchored to that taxonomy, which is implicitly endorsed by anyone using these numbers. Phase 3 is partial mitigation: it documents how much the conclusion depends on the choice of tree. A deployment in a museum or classroom context should treat the tree as configuration, not as a constant in the source code.

WikiArt is itself heavily biased toward European painting; the dataset contains 27 styles but only one (Ukiyo-e) sits outside the European-and-American canon, and East Asian ink-painting traditions that span centuries are collapsed into that single label. A retrieval system trained on this data will under-rank non-Western works for ambiguous queries. Hyperbolic geometry — which we have shown tightens local neighborhoods — could compound the effect by making those already-tight neighborhoods more confident. Anyone deploying embeddings of this kind should publish that limitation alongside the model, not bury it.

Finally, hyperbolic representations of style could in principle inform attribution or authentication of paintings. We strongly do not recommend that use without expert human review: 50–65 % top-1 accuracy over 27 well-known styles is far below the threshold any serious provenance, insurance, or legal decision should require.

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A Default style hierarchy

The default ground-truth tree, hand-built from standard art-history references and used as the primary evaluation reference throughout the paper. The complete definition lives in `scripts/hierarchy.py:STYLE_HIE`

```
Root
|-- Early_Renaissance
|   |-- Northern_Renaissance
|   `-- High_Renaissance
|       `-- Mannerism_Late_Renaissance
|           `-- Baroque
|               `-- Rococo
|                   `-- Romanticism
|                       |-- Realism
|                       |   |-- Contemporary_Realism
|                       |   `-- Impressionism
```

```
|                        |       `-- Post_Impressionism
|                        |           |-- Pointillism
|                        |           |-- Fauvism
|                        |           `-- Cubism
|                        |               |-- Analytical_Cubism
|                        |               `-- Synthetic_Cubism
|                        `-- Symbolism
|                            |-- Art_Nouveau
|                            `-- Expressionism
|                                `-- Abstract_Expressionism
|                                    |-- Action_painting
|                                    |-- Color_Field_Painting
|                                    `-- Minimalism
|-- Pop_Art
|   `-- New_Realism
|-- Ukiyo_e
`-- Naive_Art_Primitivism
```

B Sweep configuration

The sweep CSV (`notebooks/sweep_results.csv`, 150 rows) records, for every config, the full set of training hyperparameters and every evaluation metric. To reproduce:

```
python scripts/sweep.py --phase 1 --device mps   # ~14 min, 90 configs
python scripts/sweep.py --phase 2 --device mps   # ~10 min, 48 configs
python scripts/sweep.py --phase 3 --device mps   # ~3 min,  18 configs
python scripts/make_figures.py                   # regenerates docs/figures
```